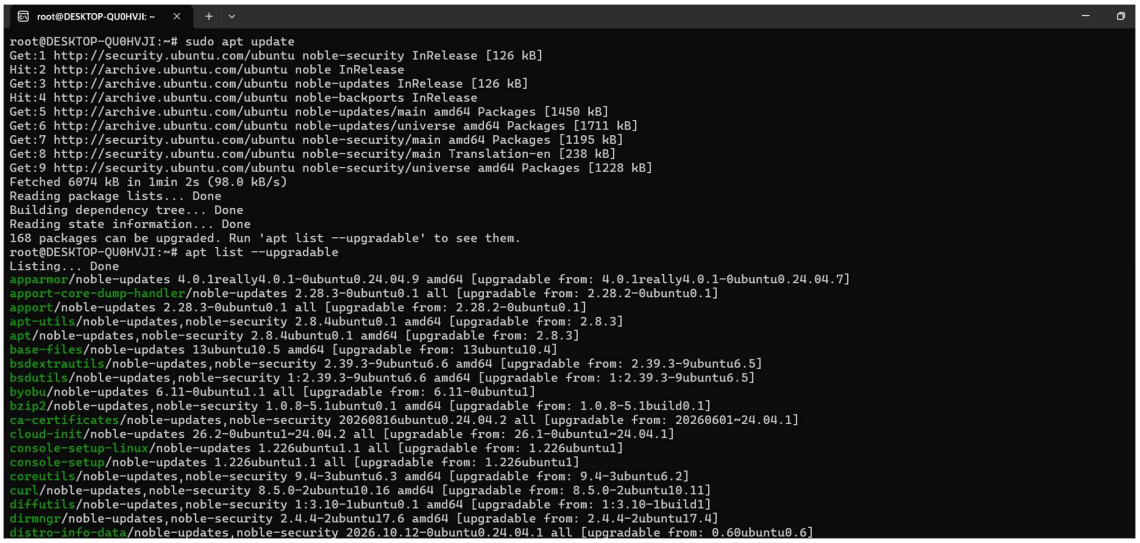
Name = HITESH SUTHAR, Roll No. = 2520090078

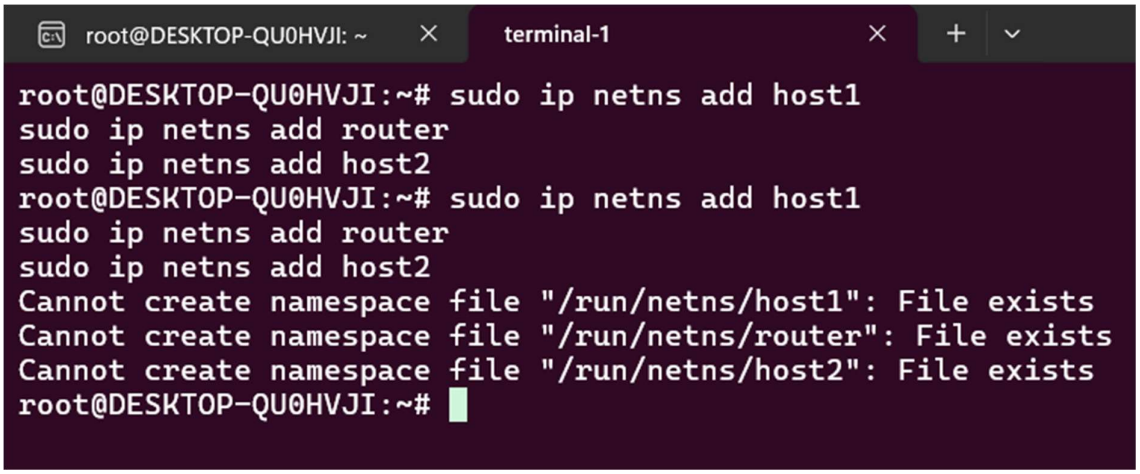
SKILL- 2

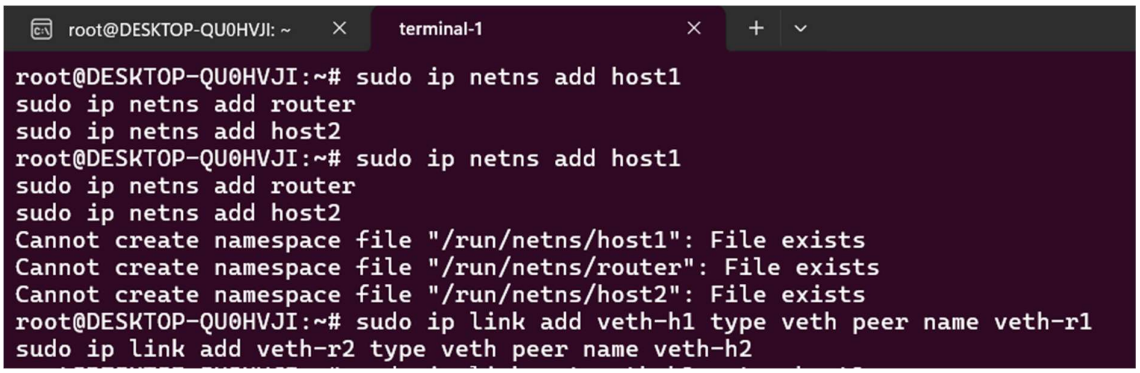
1. Install tools

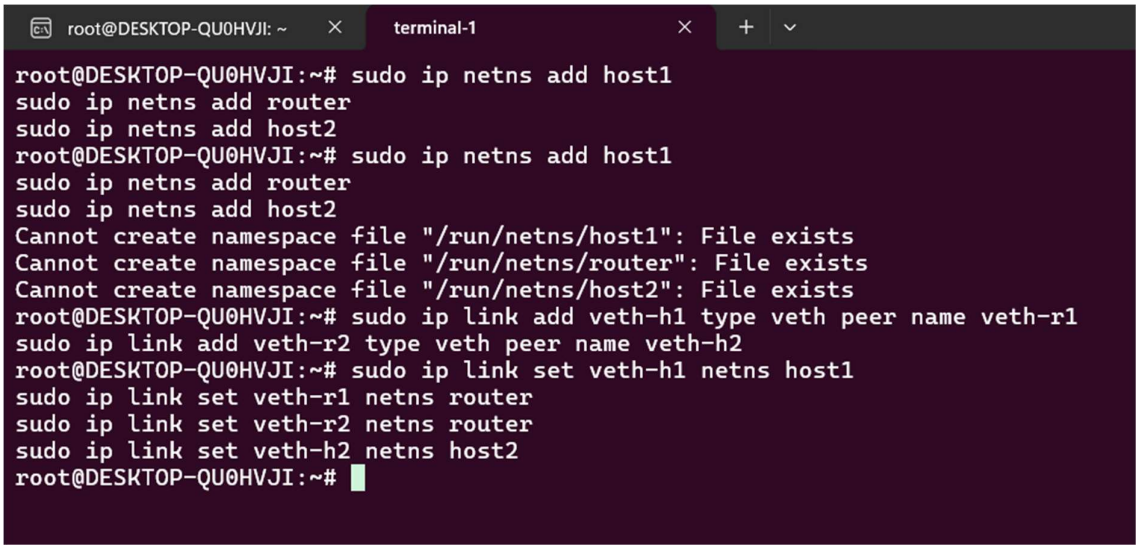


2. Build the network

Terminal-1

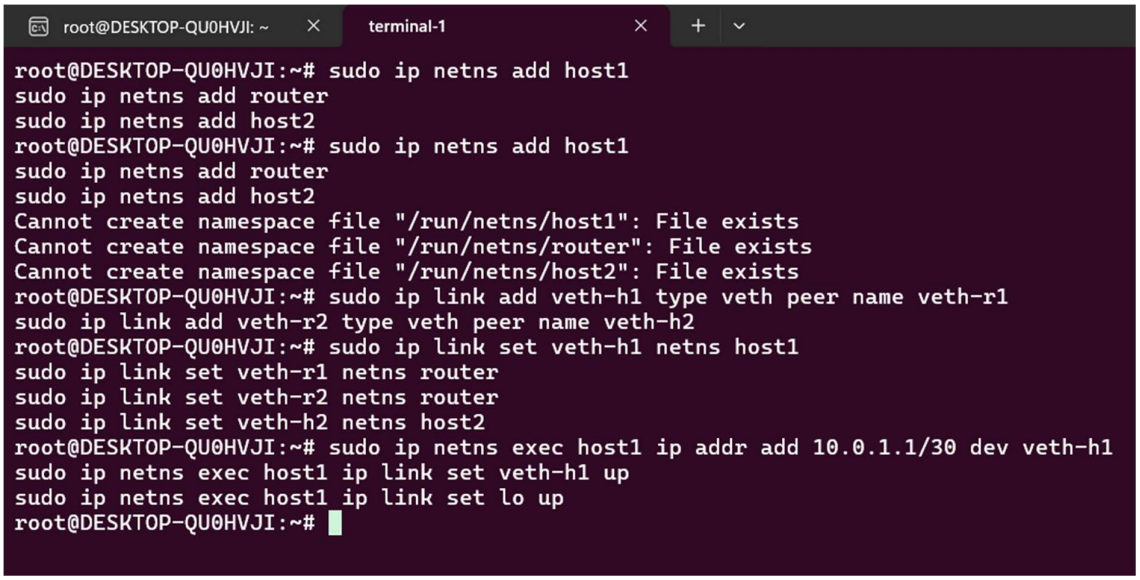




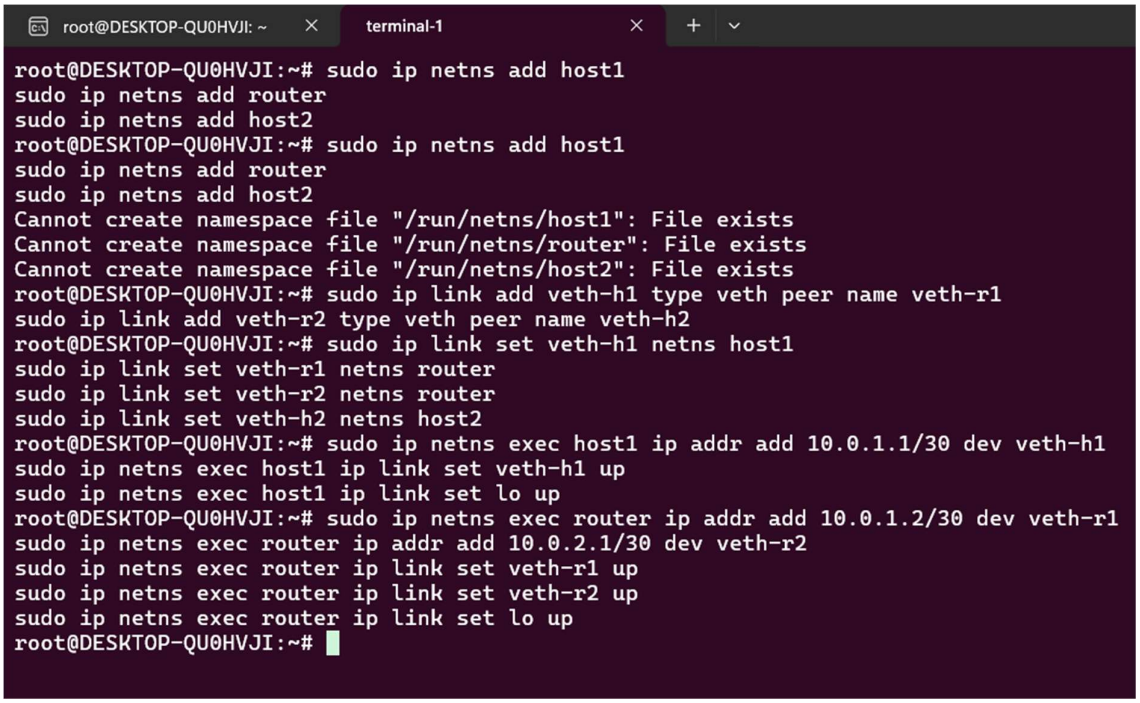


3. Configure IP addresses

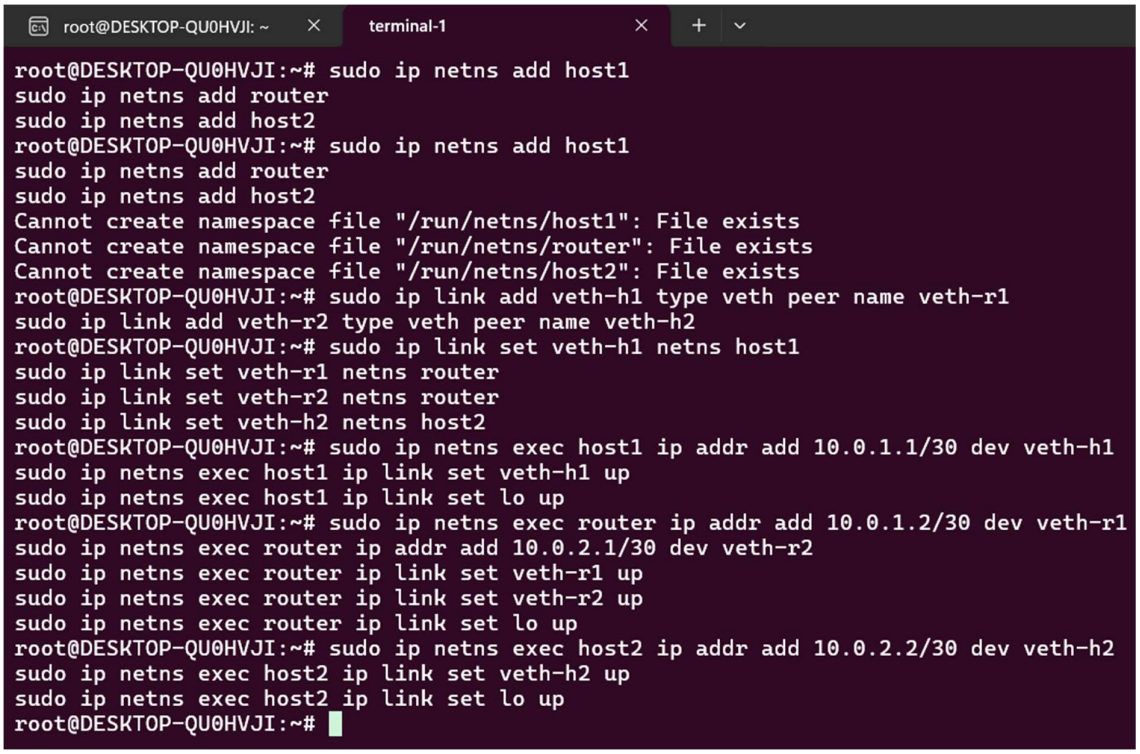
Host1:



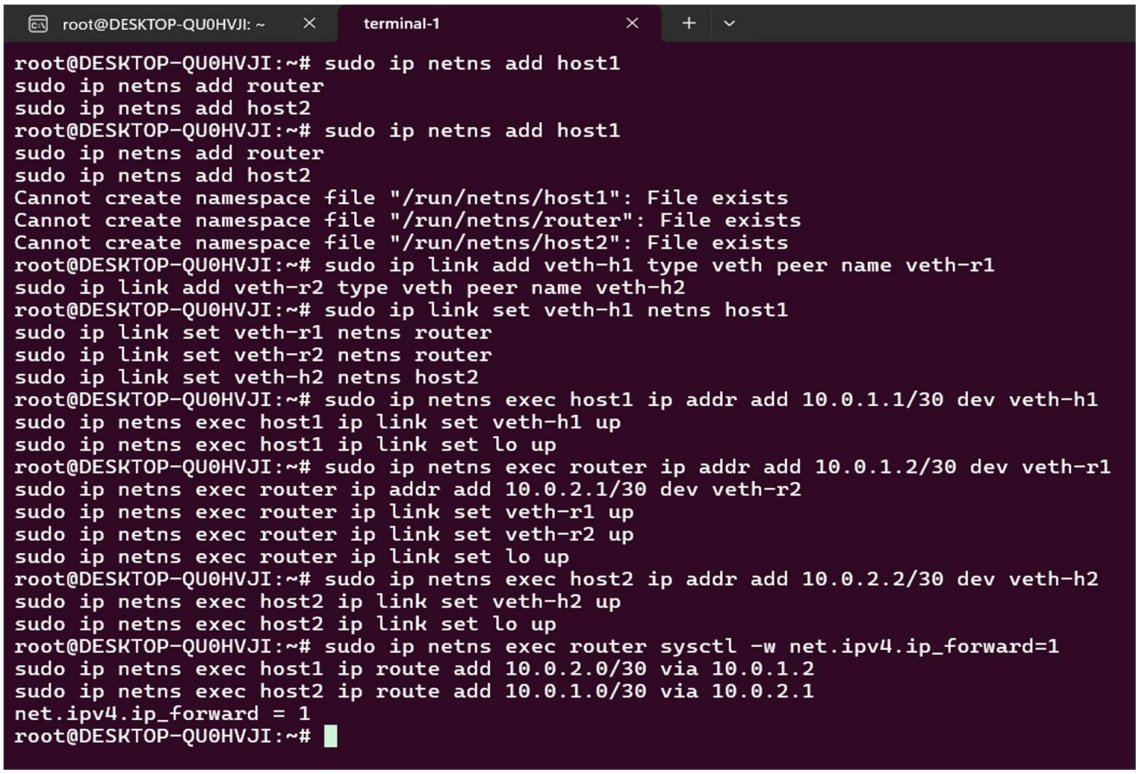
Router:


root@DESKTOP-QU0HVJI: ~
terminal-1
root@DESKTOP-QU0HVJI:~# sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
root@DESKTOP-QU0HVJI:~# sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
Cannot create namespace file "/run/netns/host1": File exists
Cannot create namespace file "/run/netns/router": File exists
Cannot create namespace file "/run/netns/host2": File exists
root@DESKTOP-QU0HVJI:~# sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
root@DESKTOP-QU0HVJI:~# sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
root@DESKTOP-QU0HVJI:~# sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
root@DESKTOP-QU0HVJI:~# sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
root@DESKTOP-QU0HVJI:~#

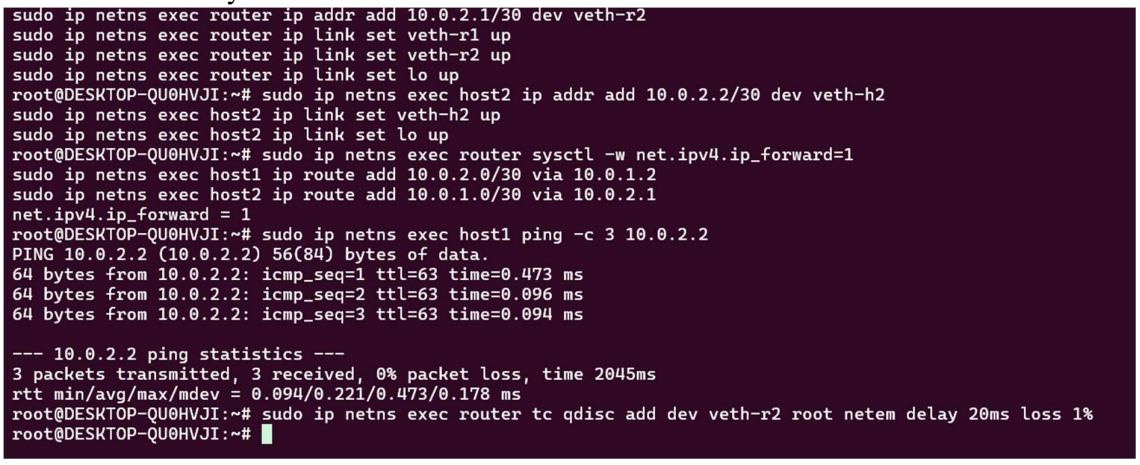
Host2:


root@DESKTOP-QU0HVJI: ~
terminal-1
root@DESKTOP-QU0HVJI:~# sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
root@DESKTOP-QU0HVJI:~# sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
Cannot create namespace file "/run/netns/host1": File exists
Cannot create namespace file "/run/netns/router": File exists
Cannot create namespace file "/run/netns/host2": File exists
root@DESKTOP-QU0HVJI:~# sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
root@DESKTOP-QU0HVJI:~# sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
root@DESKTOP-QU0HVJI:~# sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
root@DESKTOP-QU0HVJI:~# sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
root@DESKTOP-QU0HVJI:~# sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
root@DESKTOP-QU0HVJI:~#

4. Enable routing



5. Test connectivity



6. Add delay and loss

```
root@DESKTOP-QU0HVJI:~# sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
root@DESKTOP-QU0HVJI:~# sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
Cannot create namespace file "/run/netns/host1": File exists
Cannot create namespace file "/run/netns/router": File exists
Cannot create namespace file "/run/netns/host2": File exists
root@DESKTOP-QU0HVJI:~# sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
root@DESKTOP-QU0HVJI:~# sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
root@DESKTOP-QU0HVJI:~# sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
root@DESKTOP-QU0HVJI:~# sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
root@DESKTOP-QU0HVJI:~# sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
root@DESKTOP-QU0HVJI:~# sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
root@DESKTOP-QU0HVJI:~# sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.473 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.096 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.094 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2045ms
rtt min/avg/max/mdev = 0.094/0.221/0.473/0.178 ms
root@DESKTOP-QU0HVJI:~#
```

Verify:

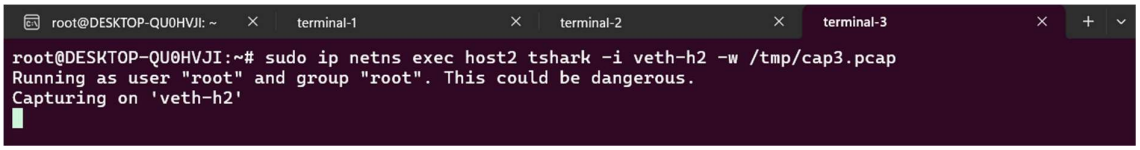
```
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
root@DESKTOP-QU0HVJI:~# sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.473 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.096 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.094 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2045ms
rtt min/avg/max/mdev = 0.094/0.221/0.473/0.178 ms
root@DESKTOP-QU0HVJI:~# sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
root@DESKTOP-QU0HVJI:~# sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1%
root@DESKTOP-QU0HVJI:~#
```

Terminal 2 — capture at router

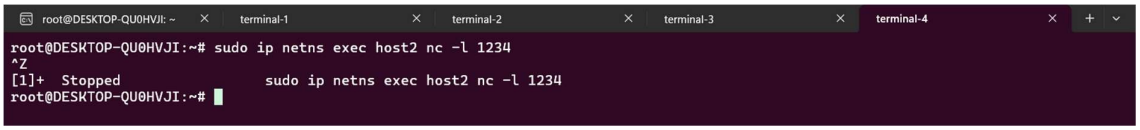
```
root@DESKTOP-QU0HVJI:~# sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
```

Terminal 3 — capture at host2

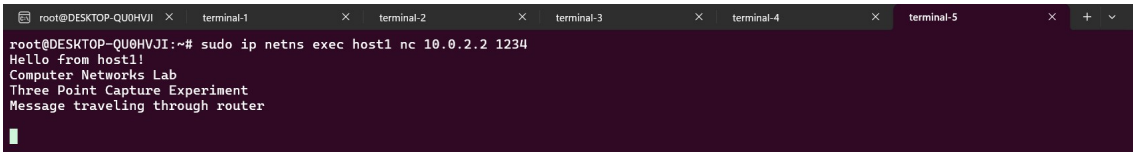


9. Send real message from host1 to host2

Terminal-4



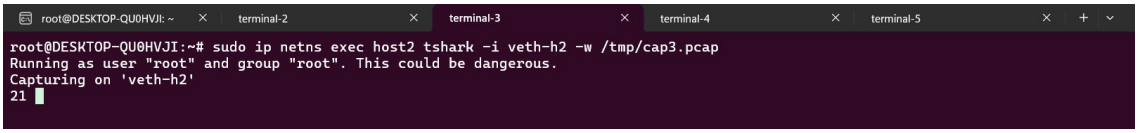
Terminal-5

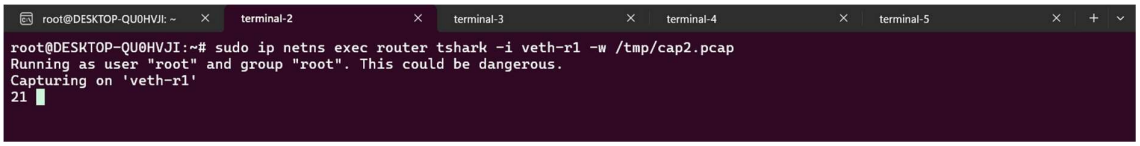


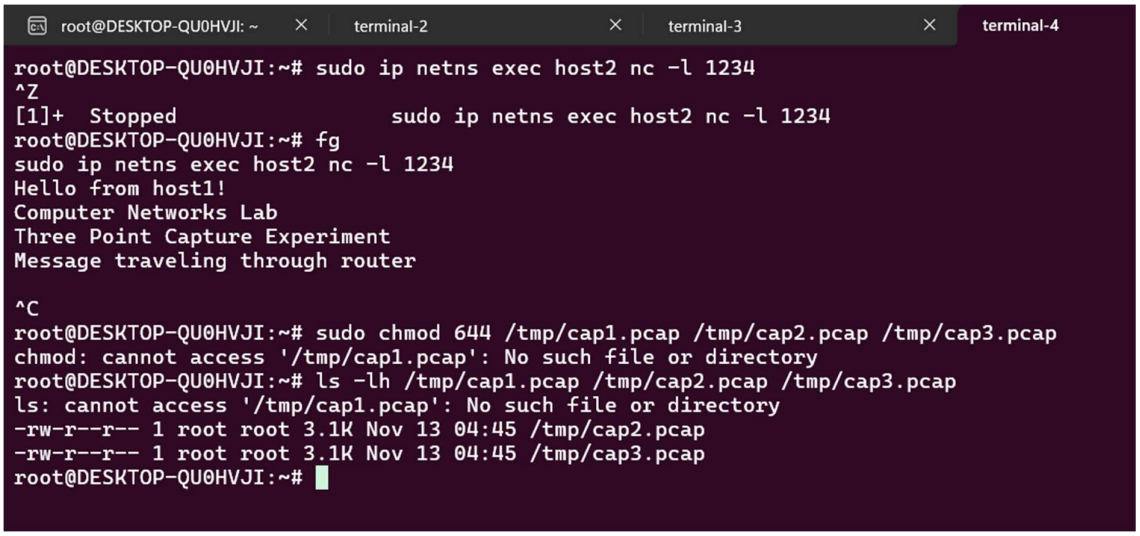
10. Terminal 4 (host2) shows your message arriving:



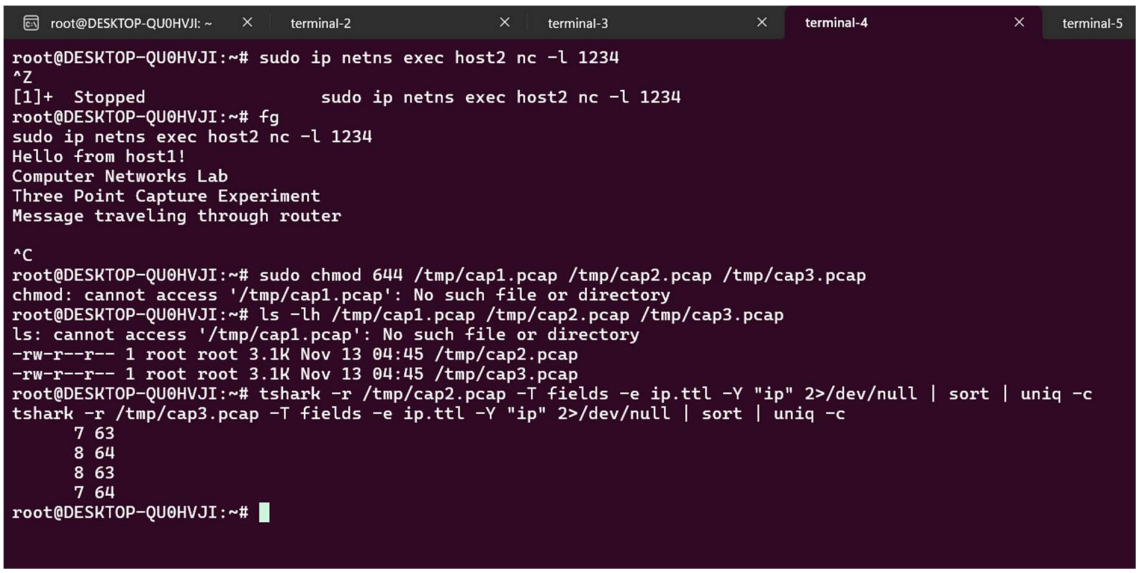
Terminal 2, 3 (tshark) show packet counts increasing:







Check TTL at all 2 points:



```
root@DESKTOP-QU0HVJI:~# tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T
fields -e frame.time_relative 2>/dev/null
tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T
fields -e frame.time_relative 2>/dev/null
tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T
fields -e frame.time_relative 2>/dev/null
Running as user "root" and group "root". This could be dangerous.
tshark: option requires an argument: T

Usage: tshark [options] ...

Capture interface:
  -i <interface>, --interface <interface>
                           name or idx of interface (def: first non-loopback)
  -f <capture filter>      packet filter in libpcap filter syntax
  -s <snaplen>, --snapshot-length <snaplen>
                           packet snapshot length (def: appropriate maximum)
  -p, --no-promiscuous-mode
                           don't capture in promiscuous mode
  -I, --monitor-mode       capture in monitor mode, if available
  -B <buffer size>, --buffer-size <buffer size>
                           size of kernel buffer (def: 2MB)
  -y <link type>, --linktype <link type>
                           link layer type (def: first appropriate)
  --time-stamp-type <type> timestamp method for interface
  -D, --list-interfaces    print list of interfaces and exit
  -L, --list-data-link-types
                           print list of link-layer types of iface and exit
  --list-time-stamp-types  print list of timestamp types for iface and exit
  --update-interval        interval between updates with new packets (def: 100ms)
```

```
Note that this can make your system less secure!
root@DESKTOP-QU0HVJI:~# sudo ip netns exec host1 mtr --report 10.0.2.2
Start: 2026-11-13T05:16:37+0000
Host:Hitesh
^C
root@DESKTOP-QU0HVJI:~#
```

15. Wireshark packet detail

